

Lauren Ostrowski

Medical Scientist Training Program
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EDUCATION

University of California San Diego

M.D./Ph.D., Neurosciences Graduate Program

2020 – present
San Diego, CA, USA

- **Supervisors:** Dr. Timothy Gentner and Dr. Vikash Gilja

Brown University

B.Sc., Computational Neuroscience with Honors

2014 – 2018
Providence, RI, USA

- Honors distinction for thesis work in developmental epilepsy

RESEARCH POSITIONS

Gentner and Translational Neuroengineering Labs

PhD Student

2022 – present
UCSD, San Diego, CA, USA

- Studying multi-regional neural population dynamics underlying the biomechanical control of learned vocal motor sequences in the songbird model system for speech production
 - *Developed novel neurosurgical techniques, custom software, and custom 3D-printed hardware for Neuropixels 2.0 probes, obtaining large-scale recordings of simultaneous single-unit activity in songbird HVC and RA across hundreds of song renditions*
- **Research residency at the Neural Interfacing Lab (PI: Christian Herff), Maastricht University, The Netherlands (March – July 2025, funded by the Merkin Graduate Fellowship)** investigating the neural mechanisms of bilingual speech production and low-dimensional dynamical regimes of language selectivity in intracranial recordings in Dutch-English speakers
 - *Designed two complementary paradigms, picture naming and unscripted dialogue with synthetic interlocutors, to elicit bilingual speech under ecologically valid conditions*
- Investigating temporal dynamics of linguistic and paralinguistic feature processing during speech perception across native and non-native languages using magnetoencephalography (MEG) and time-resolved neural decoding

Halgren Lab

Rotation Student

2021
UCSD, San Diego, CA, USA

- Explored neural activity in the human brain during speech perception using intraoperative data from high-density micro-ECoG electrodes placed over the superior temporal gyrus
- Using custom MATLAB-based analysis, found that phoneme-specific high-frequency oscillations phase-locked and co-occurred when phonemes were bound in consonant-vowel pairs

Computational Neural DNA Dynamics Lab

Rotation Student

2020
UCSD, San Diego, CA, USA

- Studied the frequency of atypical allele counts in healthy brain tissue using cytosine DNA methylation and chromatin profiles of post-mortem human frontal cortex tissue
- Developed and validated a pipeline in Python to detect partial aneuploidy at the single cell level

Angelman Research Team at Biogen

Data Analyst

2019 – 2020
Cambridge, MA, USA

- Analysis of EEG data in Angelman Syndrome and design of a robust, non-invasive EEG biomarker for Phase 1 clinical trial
- EEG findings eliminated the need for serial lumbar punctures in deployment of clinical trial

BrainGate Consortium

2018 – 2019

Research Assistant

Brown University and VA Medical Center, Providence, RI, USA

- Engineered software for neural signal processing with multi-institutional distribution
- Analyzed EEG activity in patients with recent stroke for future implementation in brain-computer interface

Chu Lab

2016 – 2018

Research Assistant

Massachusetts General Hospital, Boston, MA, USA

- Combined non-invasive physiological imaging techniques and behavioral data to identify biomarkers associated with seizure activity and neuropsychological impairments in developmental epilepsy
- Found correlation between white matter microstructure in language regions and comorbid language impairments
- Modeled whole-brain structural connectivity and analyzed network responses to electrical stimulation

The Feinstein Institute for Medical Research

2015 – 2016

Research Assistant

New Hyde Park, NY, USA

- Analyzed efficacy and interventional design of long-term health program to combat obesity and type 2 diabetes

GRANTS AND AWARDS

2026	UCSD Neurosciences Graduate Access and Enrichment Grant
2024	Sanford Institute for Empathy and Compassion Research Fellowship
2024	UCSD Diversity in Neuroscience Enrichment Grant
2023	Merkin Graduate Fellowship <i>Supported research residency at Maastricht University, The Netherlands</i>
2023	ACM SIGHPC Computational & Data Science Fellowship
2018	Sigma Xi Scientific Research Honor Society
2018	Bachelor of Science in Neuroscience with Honors Designation
2017	Presidential LINK Award, Brown University
2014	U.S. Presidential Scholar semi-finalist
2014	New Hampshire Red Sox Service Scholarship
2014	Milford Rotary Scholarship
2014	Joshua Sayvon Scholarship

PUBLICATIONS

Preprints

- [1] **Ostrowski LM**, Méndez JM, Tostado-Marcos P, Cooper BG, Gentner TQ. Automated inference of respiratory and syringeal biomechanical trajectories from birdsong acoustics. *bioRxiv* 2026.08.03.742634.
- [2] Weger P, Ottenhoff MC, Verwoert M, Gimple SV, **Ostrowski LM**, . . . , Kubben P, Herff C. Insular error network enables self-correcting intracranial brain-computer interface. *bioRxiv* 2025.11.17.688824.

Peer-reviewed articles

- [3] Tostado-Marcos P, Arneodo EM, **Ostrowski LM**, Brown DE II, . . . , Gentner TQ, Gilja V. Population dynamics in songbird RA and HVC during learned motor-vocal behavior. *J Neurosci* 2026;e0580252026.
- [4] Hardstone R, **Ostrowski LM**, Dusang AN, López-Larraz E, . . . , Lin DJ. Extension of voxel-based lesion mapping to multidimensional neurophysiological data. *Sci Rep* 2025;15:41488.
- [5] **Ostrowski LM**, Chinappen DM, Stoyell SM, Song DY, Ross EE, . . . , Chu CJ. Children with Rolandic epilepsy have micro- and macrostructural abnormalities in white matter constituting networks necessary for language function. *Epilepsy Behav* 2023;144:109254.
- [6] Chinappen DM, **Ostrowski LM**, Spencer ER, Kwon H, Kramer MA, . . . , Chu CJ. Decreased thalamocortical connectivity in resolved Rolandic epilepsy. *Clin Neurophysiol* 2023;153:21-27.
- [7] Spencer ER, Shi W, Komorowski RW, Gilbert JP, **Ostrowski LM**, . . . , Chu CJ. Longitudinal EEG model detects antisense oligonucleotide treatment effect and increased UBE3A in Angelman syndrome. *Brain Commun* 2022;4(3):fcac106.

- [8] **Ostrowski LM**, Spencer ER, Bird LM, Thibert R, Komorowski RW, Kramer MA, Chu CJ. Delta power robustly predicts cognitive function in Angelman syndrome. *Ann Clin Transl Neurol* 2021;8(7):1433-1445.
- [9] Crocker B, **Ostrowski LM**, Williams ZM, Dougherty DD, Eskandar EN, . . . , Cash SS, Paulk AC. Local and distant responses to single pulse electrical stimulation reflect different forms of connectivity. *Neuroimage* 2021;237:118094.
- [10] Kramer MA, Stoyell SM, Chinappen DM, **Ostrowski LM**, Spencer ER, . . . , Chu CJ. Focal sleep spindle deficits reveal focal thalamocortical dysfunction and predict cognitive deficits in sleep activated developmental epilepsy. *J Neurosci* 2021;41(8):1816-1829.
- [11] Stoyell SM, Baxter BS, McLaren J, Kwon H, Chinappen DM, **Ostrowski LM**, . . . , Chu CJ. Diazepam induced sleep spindle increase correlates with cognitive recovery in a child with epileptic encephalopathy. *BMC Neurol* 2021;21(1):355.
- [12] Thorn EL, **Ostrowski LM**, Chinappen DM, Jing J, Westover MB, . . . , Chu CJ. Persistent abnormalities in Rolandic thalamocortical white matter circuits in childhood epilepsy with centrotemporal spikes. *Epilepsia* 2020;61(11):2500-2508.
- [13] **Ostrowski LM**, Song DY, Thorn EL, Ross EE, Stoyell SM, . . . , Stufflebeam SM, Chu CJ. Dysmature superficial white matter microstructure in developmental focal epilepsy. *Brain Commun* 2019;1(1):fcz002.
- [14] **Ostrowski LM**, Speiser PW, Accacha S, Altshuler L, Fennoy I, . . . , Rosenbaum M. Demographics and anthropometrics impact benefits of health intervention: Data from the Reduce Obesity and Diabetes project. *Obes Sci Pract* 2019;5(1):46-58.
- [15] Kramer MA, **Ostrowski LM**, Song DY, Thorn EL, Stoyell SM, . . . , Chu CJ. Scalp recorded spike ripples predict seizure risk in childhood epilepsy better than spikes. *Brain* 2019;142(5):1296-1309.
- [16] Song DY, Stoyell SM, Ross EE, **Ostrowski LM**, Thorn EL, . . . , Chu CJ. Beta oscillations in the sensorimotor cortex correlate with disease and remission in benign epilepsy with centrotemporal spikes. *Brain Behav* 2019;9(3):e01237.

TEACHING

2023 – present	Founder and Workshop Leader, Brain Machine Interface AI/ML Hackathon, <i>UCSD</i>
2023 – 2025	Discussion Leader, Neurosciences Graduate Program Journal Club, <i>UCSD</i>
2024	Teaching Assistant, Minor Proposition, <i>UCSD</i>
2022	Guest Lecturer, <i>UCSD School of Medicine</i> <i>Lecture: "Health Care for People Experiencing Homelessness in San Diego"</i>
2016	Teaching Assistant, General Chemistry, <i>Brown University</i>

MENTORSHIP

2022 – present	Undergraduate Research Mentoring
2020 – present	UCSD Pre-Med Mentoring Program

SERVICE

2025 – present	Teacher and Curriculum Developer: Las Colinas Detention Center <i>Prepare and teach classes at grant-funded science education program for incarcerated individuals in Santee, CA; in English and Spanish</i>
2022 – 2025	Neurosciences Graduate Program Committees: Chair, NGP Student Wellness Committee; Member, NGP Diversity Committee
2021 – 2022	Chapter President: Association of Women Surgeons, UCSD Student Chapter
2021 – 2022	Leadership & Clinical Volunteer: Homeless Health Initiative & Safe Haven Clinic
2021 – 2022	Public Health Advisory Council, Climate Action Campaign <i>Social Media Manager, patient health advocate at SD County Board of Supervisors meetings</i>
2021 – 2022	Operation Harm Reduction <i>Volunteer distributing Narcan, clean needles, and first aid supplies</i>
2020 – present	Medical Scientist Training Program Committees: Member, MSTP Diversity Committee; Member, MSTP LGBTQ+ Health Committee

2020 – 2022	Executive Committee: LGBTQ+ Pharmacy & Medicine <i>Organized SoCal LGBTQ+ Conference, Trans Week of Visibility, UCSD Rainbow Pin Ceremony</i>
2020 – 2022	Executive Committee: Border Health Project <i>Organized mobile health clinics, supply drives for migrant farmworkers</i>
2020 – 2022	Anti-Racism Coalition at UCSD <i>Policy & Admissions Subcommittees: combating ICE in healthcare, microaggression reporting</i>
2020 – 2021	Clinic Manager: UCSD Women's Free Clinic <i>Managed clinic operations and planned contraceptive counseling services</i>
2017 – 2019	Clinical Volunteer: Clínica Esperanza / Hope Clinic, Providence, RI
2017 – 2018	President: Women in Science and Engineering at Brown
2017 – 2018	Mentor: The Adolescent Leadership Council, Providence, RI <i>Adolescent chronic illness support group</i>
2015 – 2018	Peer Advisor: The Meiklejohn Peer Advising Program, Brown University

PRESENTATIONS

- [1] Society for the Neurobiology of Language 2026, Geneva, Switzerland. (September 29 – October 2, 2026). Poster: "Asymmetric dynamical regimes underlie bilingual language control in intracranial recordings from Dutch-English speakers." **L. Ostrowski**, et al.
- [2] Computational and Systems Neuroscience (COSYNE) 2026, Lisbon, Portugal. (March 12 – 17, 2026). Poster: "Transient, behavior-dependent coupling of neural population dynamics during learned motor sequences." J. Gorman*, **L. Ostrowski***, et al. [*equal contribution]
- [3] Society for Neuroscience 2025, Chicago, IL, USA. (November 15 – 18, 2025). Poster: "Cortical temporal dynamics of speech perception: Dissociating linguistic and paralinguistic elements." J. Xing, **L. Ostrowski**, et al.
- [4] Society for Neuroscience 2025, Chicago, IL, USA. (November 15 – 18, 2025). Poster: "Latent premotor neural dynamics reveal species-specific vocal strategies in songbirds." J. Gorman, **L. Ostrowski**, et al.
- [5] Society for Neuroscience 2025, Chicago, IL, USA. (November 15 – 18, 2025). Poster: "Species differences in real-time sensorimotor error processing during vocalization." L. Stanwicks, **L. Ostrowski**, et al.
- [6] American Society of Neuroradiology 2023, Los Angeles, CA, USA. (March 15, 2023). Poster: "Linking post-stroke neurophysiology to neuroanatomy: Novel method to extend voxel-lesion mapping to multi-dimensional EEG data." R. Hardstone, **L. Ostrowski**, et al.
- [7] American Society of Neuroradiology 2022, Boston, MA, USA. (April 1, 2022). Poster: "Persistent asymmetry of aperiodic resting-state neural activity in cortical and subcortical strokes." R. Hardstone, **L. Ostrowski**, et al.
- [8] Society for Neuroscience 2018, San Diego, CA, USA. (November 3 – 7, 2018). Poster: "Focal abnormalities in white matter microstructure correspond to neuropsychological comorbidities in BECTS." **L. Ostrowski**, et al.
- [9] American Epilepsy Society 2017, Washington DC, USA. (December 1 – 5, 2017). Poster: "Dysmature white matter microstructure in developmental epilepsy." **L. Ostrowski**, et al.
- [10] Endocrine Society 2016, Boston, MA, USA. (April 1 – 4, 2016); Obesity Society 2017, Washington, DC, USA. (October 29 – November 2, 2017). Poster: "Demographics and anthropometrics impact benefits of health intervention: Data from the ROAD Project." **L. Ostrowski**, et al.

SKILLS

Programming languages: Python, MATLAB, C++

Scientific computing & ML: PyTorch, TensorFlow/Keras, NumPy, SciPy, Scikit-learn, statsmodels, Pandas, generalized linear models, dimensionality reduction, neural decoding, dynamical-systems analysis

Neurophysiology & signal processing: spike sorting (Kilosort), SpikeGLX/Open Ephys, NWB, MNE-Python, spectral/signal processing, DTI/tractography (FSL, MRtrix, FreeSurfer, Probtrackx)

Hardware & fabrication: 3D printing, Solidworks, Onshape

Tools: Git, Linux/Unix systems, HPC/cluster computing (SLURM), Containers (Docker), AWS, Adobe Illustrator, Adobe Photoshop, LaTeX

Languages: English, Spanish